

Fig. 3: Block diagram of RFID proximity card

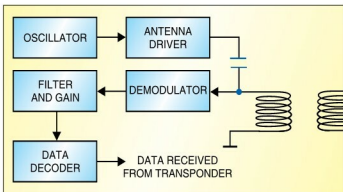


Fig. 4: Block diagram of RFID transceiver



Fig. 5: Prototype of a microcontroller-based RFID building access control system (Courtesy: www.cytron.com.my)

paper 'Communication by Means of Reflected Power' in 1948.

RFID applications

There are many kinds of RFID systems used in different applications and settings. These systems have different power sources, operating frequencies and functionality.

At the core, RFID systems consist of small transponders (tags) attached to physical objects. When queried by RFID systems that transmit data via radio signals do not have the same performance limitations as optical systems—according to *Handbook of Technology Management*.

RFID system requisites

Common applications of RFID systems include tracking and access control. Robert

Weiss, in his book *RFID: Principles and Application*, says that a standard RFID system consists of at least three core components:

1. RFID tags (transponders), which carry object-identifying data
2. RFID readers (transceivers), which read and write tag data
3. Databases, which associate

tag-identifying data with arbitrary records

The basic function of an RFID system is identification. When queried by a reader, tags return an identifier that may be used to retrieve other data records. The tag is attached to the object to be identified.

Typically, the tags consist of an antenna and an electronic circuit. Often, these carry no on-board battery (passive tags) and must harvest all energy from the RF signal. Newer tags tend to implement identification functionality on a chip (IC) that provides computation and storage. In the manufacturing process, this tiny chip is attached to an antenna before being packaged in a form factor (glass capsule or foil) inlay, which is integrated into the finished product.

The reader communicates with the tag through a radio-frequency channel to obtain the identifying information. Depending on the type of tag, this communication may be a simple ping or a more complex multi-round protocol.

Readers come in many forms, operate on many different frequencies and may offer a wide range of functionality. These may have their own processing power and internal storage, and may offer network connectivity. Readers might be a simple conduit to an external system or store all relevant data locally. Databases are assumed to have a secure connection to readers.

Presence sensing and building access control

RFID proximity cards (Prox Cards) are widely used for presence sensing and building access control throughout the world. These are passive tags with no on-board battery, relying solely on a reader to operate. Usually, passive tags do not have their own power source or the ability to initiate communication. These obtain energy from incoming RF signals by induction. Often, the reader energises the tag through the RF channel.

While passive tags have the short-